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PAPER_939: Centaurus A Jet Power Curves

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 213
Source: blazar_{jet_power_curves_extended}.py (CentaurusAJetPowerCurves) **Calculator:** CentaurusAJetPowerCurvesCalc (CP4 #523) **CVW:** v2.0.0 compliant

Abstract

We compute numerical jet power curves for Centaurus A (NGC 5128), the nearest radio galaxy at $d \approx 3.8$ Mpc. Using the Blandford-Znajek mechanism with UQFF phonon linewidth modulation, we derive power enhancement factors at three linewidths $\Gamma = 0.05, 0.10, 0.30$ THz. The SMBH parameters are $M_{\text{BH}} = 5.5 \times 10^7 M_{\odot}$, $a = 0.70$, $B = 3000$ T, and $A_{\text{jet}} = 0.95$, yielding enhancements of $2.6 \times / 2.1 \times / 1.4 \times$ respectively.

1. System Parameters

Parameter	Value	Source
M_{BH}	$5.5 \times 10^7 M_{\odot}$	EHT/VLBI
Spin a	0.70	Jet morphology fits
B	3000 T	Faraday rotation
A_{jet}	0.95	Phonon coupling
Distance	3.8 Mpc	Tully-Fisher

2. Blandford-Znajek Power

$$P_{\text{BZ}} = \frac{B^2}{8\pi} \left(\frac{r_H}{c} \right)^2 a^2 c$$

where $r_H = \frac{r_S}{2} (1 + \sqrt{1 - a^2})$ and $r_S = 2GM/c^2$.

3. UQFF Jet Modulation

$$M_{\text{jet}}(\Gamma) = 1 + A_{\text{jet}} \exp! \left(-\frac{(\Gamma - \Gamma_0)^2}{2\sigma_{\text{Gamma}}^2} \right)$$

$$P_{\text{jet}}(\Gamma) = P_{\text{BZ}} \cdot (1 + M_{\text{jet}}(\Gamma))$$

Γ (THz)	Enhancement
0.05	$2.6\times$
0.10	$2.1\times$
0.30	$1.4\times$

4. Source Data

- **File:** blazar_{jet_power_curves_extended}.py
- **Session:** 213
- **CP4 Class:** CentaurusAJetPowerCurvesCalc (#523)

References

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Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{\{U_Bi_i\}}$ jet modulation curves and PAPER_1048 for phonon-corrected $M\text{-}\sigma$ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

$M\text{-}\sigma$ correction (PAPER_1048): The phonon-corrected $M\text{-}\sigma$ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
String sector coupling	$[SSq]$	0.57	BH dynamics

Constant	Symbol	Value	Validation Domain
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	Phonon resonance
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37} \text{ J/m}^3$	Fundamental

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Phonon frequency ω_{SCm}	$2\pi \times 1.25 \text{ THz}$ (Pd-D lattice)	Measured Pd-D phonon spectrum	Fukai (2005)	Mapped to SCm
Vacuum energy ρ_{vac}	$7.09 \times 10^{-37} \text{ kg/m}^3$	$\rho_{vac} \sim 10^{-29} \text{ g/cm}^3$	Planck 2018	Novel SCm scale
Fine structure α	UQFF reproduces via U_{g1} dipole	1/137.036	PDG 2024	99.9%

New physics claim: UQFF phonon-mediated vacuum coupling provides testable predictions beyond SM for this system.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: SCm-phonon (lattice resonance)

§A.2 Lagrangian Density

$$\mathcal{L}_{SCm_phonon} = \sum_{i=1}^{26} [U_{g,i} + U_{m,i} + U_{A,i} - U_{b,i}] \cdot S_{26}([SSq]) \cdot \Phi_{1.25\text{THz}}(\omega, \Gamma)$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_m u \frac{\partial \mathcal{L}}{\partial (\partial_m u \phi)} = 0 \implies F_{U,Bi_i} = -\nabla U_{\text{eff}} + \Phi \cdot S_{26} \cdot E_{\text{net}}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms $\rightarrow$ SCm vacuum $\rightarrow$ phonon ω_{SCm} $\rightarrow$ lattice resonance $\rightarrow F_{U,Bi_i}$ unified force $\rightarrow$ observational prediction

§B VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

$$\text{VDS} = \rho_{SCm} \cdot S_{26} \cdot \Phi_{1.25\text{THz}} / \Phi_0$$

VDS sub-ratio: 0.1

§B.2 Dipole Vortex Primes (DVP)

DVP prime: 2 (sub-threshold)

§B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale: 10^4 yr

§B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.1	Confirmed
κ decay	5×10^{-4}	Confirmed
$[SSq]$	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1009	3C273 AGN F_{U_Bi_i} Jet Modulation
PAPER_1010	TON618 AGN F_{U_Bi_i} Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy

10 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

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